

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO4-AT2 - Laboratory Performance | Programming in Java

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Course Code	CSA0904
Course Name	Programming in Java
Assignment	CO4-AT2 - Laboratory Performance (Question 2)

QUESTION 2: GUI USING AWT CONTROLS (REGISTRATION FORM)

Problem Statement:

Create a GUI using AWT controls (Button, Label, TextField, Checkbox, Choice). Perform simple operations such as registration form input and display the entered data.

SOURCE CODE

```
import java.awt.*;
import java.awt.event.*;

// Main Class at the Top
public class StudentRegistrationAWT extends Frame implements ActionListener {

    // AWT Controls
    private Label lblTitle, lblName, lblEmail, lblGender, lblCourse, lblInterests, lblOutput;
    private TextField txtName, txtEmail;
    private CheckboxGroup genderGroup;
    private Checkbox chkMale, chkFemale;
    private Choice chCourse;
    private Checkbox chkCoding, chkSports, chkMusic;
    private Button btnSubmit, btnClear;
    private TextArea txtAreaDisplay;

    public StudentRegistrationAWT() {
        // Frame Configuration
        setTitle("Student Registration Form - AWT Controls");
        setSize(520, 680);
        setBackground(new Color(245, 247, 250));
        setLayout(new FlowLayout(FlowLayout.CENTER, 15, 12));
        setLocationRelativeTo(null);

        // Header
        lblTitle = new Label("STUDENT REGISTRATION FORM", Label.CENTER);
        lblTitle.setFont(new Font("Arial", Font.BOLD, 16));
        lblTitle.setForeground(new Color(30, 60, 114));
        add(lblTitle);

        // Name Field
        lblName = new Label("Full Name:");
        lblName.setFont(new Font("Arial", Font.BOLD, 12));
        add(lblName);
```

```
txtName = new TextField(30);
add(txtName);

// Email Field
lblEmail = new Label("Email Address:");
lblEmail.setFont(new Font("Arial", Font.BOLD, 12));
add(lblEmail);
txtEmail = new TextField(30);
add(txtEmail);

// Gender (Checkbox with CheckboxGroup -> Radio Buttons)
lblGender = new Label("Gender:");
lblGender.setFont(new Font("Arial", Font.BOLD, 12));
add(lblGender);
genderGroup = new CheckboxGroup();
chkMale = new Checkbox("Male", genderGroup, true);
chkFemale = new Checkbox("Female", genderGroup, false);
add(chkMale);
add(chkFemale);

// Course Selection (Choice dropdown)
lblCourse = new Label("Select Course:");
lblCourse.setFont(new Font("Arial", Font.BOLD, 12));
add(lblCourse);
chCourse = new Choice();
chCourse.add("B.Tech Computer Science");
chCourse.add("B.Tech Information Technology");
chCourse.add("B.Tech AI & Data Science");
chCourse.add("B.Tech Electronics & Comm.");
add(chCourse);

// Areas of Interest (Multiple Checkboxes)
lblInterests = new Label("Areas of Interest:");
lblInterests.setFont(new Font("Arial", Font.BOLD, 12));
add(lblInterests);
chkCoding = new Checkbox("Coding / Dev");
chkSports = new Checkbox("Sports");
chkMusic = new Checkbox("Music & Arts");
add(chkCoding);
add(chkSports);
add(chkMusic);

// Action Buttons
btnSubmit = new Button("Submit Registration");
btnSubmit.setBackground(new Color(41, 128, 185));
btnSubmit.setForeground(Color.WHITE);
btnSubmit.setFont(new Font("Arial", Font.BOLD, 12));
btnSubmit.addActionListener(this);
add(btnSubmit);

btnClear = new Button("Clear");
btnClear.setBackground(new Color(231, 76, 60));
btnClear.setForeground(Color.WHITE);
btnClear.setFont(new Font("Arial", Font.BOLD, 12));
btnClear.addActionListener(this);
add(btnClear);

// Output Display Area
lblOutput = new Label("--- Registered Information ---", Label.CENTER);
lblOutput.setFont(new Font("Arial", Font.BOLD, 12));
add(lblOutput);

txtAreaDisplay = new TextArea(10, 48);
txtAreaDisplay.setEditable(false);
txtAreaDisplay.setFont(new Font("Monospaced", Font.PLAIN, 12));
txtAreaDisplay.setBackground(Color.WHITE);
```

```

add(txtAreaDisplay);

// Window Closing Listener
addWindowListener(new WindowAdapter() {
    @Override
    public void windowClosing(WindowEvent e) {
        dispose();
        System.exit(0);
    }
});
}

@Override
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == btnSubmit) {
        String name = txtName.getText().trim();
        String email = txtEmail.getText().trim();
        String gender = genderGroup.getSelectedCheckbox().getLabel();
        String course = chCourse.getSelectedItem();

        // Collect selected interests
        StringBuilder interests = new StringBuilder();
        if (chkCoding.getState()) interests.append("Coding, ");
        if (chkSports.getState()) interests.append("Sports, ");
        if (chkMusic.getState()) interests.append("Music, ");

        if (interests.length() > 0) {
            interests.setLength(interests.length() - 2); // remove trailing comma
        } else {
            interests.append("None selected");
        }

        // Input Validation
        if (name.isEmpty() || email.isEmpty()) {
            txtAreaDisplay.setText("Error: Please fill in all mandatory fields (Name and Email).");
            return;
        }

        // Format and Display Data
        StringBuilder result = new StringBuilder();
        result.append("=====\n");
        result.append("          REGISTRATION SUCCESSFUL          \n");
        result.append("=====\n");
        result.append("Full Name      : ").append(name).append("\n");
        result.append("Email ID       : ").append(email).append("\n");
        result.append("Gender         : ").append(gender).append("\n");
        result.append("Course         : ").append(course).append("\n");
        result.append("Interests      : ").append(interests.toString()).append("\n");
        result.append("=====");

        txtAreaDisplay.setText(result.toString());

    } else if (e.getSource() == btnClear) {
        txtName.setText("");
        txtEmail.setText("");
        genderGroup.setSelectedCheckbox(chkMale);
        chCourse.select(0);
        chkCoding.setState(false);
        chkSports.setState(false);
        chkMusic.setState(false);
        txtAreaDisplay.setText("");
        txtName.requestFocus();
    }
}

public static void main(String[] args) {

```

```
        new StudentRegistrationAWT().setVisible(true);  
    }  
}
```

OUTPUT SCREENSHOT

